1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with favorable immune architecture still fail to achieve better clinical outcomes?

Working Hypothesis

Primary hypothesis: In muscle-invasive bladder cancer (MIBC), the prognostic value of tumor-infiltrating lymphocyte (TIL) density for progression-free interval (PFI) is governed by tumor PD-L1 (CD274) expression. Specifically, TIL-high tumors with CD274-high status will have shorter PFI than TIL-high tumors with CD274-low status, due to PD-L1–mediated suppression of CD8A+ cytotoxic T-cell activity; conversely, in CD274-low tumors, high TIL density will associate with longer PFI, reflecting effective anti-tumor immunity. Secondary modifiers: loss of HLA class I antigen presentation (e.g., B2M/HLA-A/B/C disruption) and SARIFA (tumor-adipocyte/stromal interface associated immune exclusion) will further blunt or modulate the protective effect of TILs, potentially attenuating the favorable impact of TIL-high status even when CD274 is low. Overall, the interaction between TIL density and CD274, modulated by HLA class I status and SARIFA, will provide a more accurate, mechanistic predictor of PFI than TILs alone in MIBC.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CD274, CD8A
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD274: CD274) which provides a mechanistic suppressor of T-cell function; I selected these because they directly explain how local immune quantity (TILs) can be decoupled from immune quality/function.

CD8A: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	HRR (hemoglobin, RDW)	Supporting modifier from knowledge graph

Secondary	HLA-A / HLA-B / HLA-C (class I antigen presentation)	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	SARIFA (tumor–adipocyte/stromal interface)	Supporting modifier from knowledge graph
Secondary	PTGDS (prostaglandin signaling axis)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Immunoscore-type measures reflect immune density but can be misleading if functional suppression via PD-L1 undermines anti-tumor activity;

Prediction 2: PD-L1 expression on tumor cells or within the microenvironment has been associated with immune evasion and variable prognosis across bladder cancer cohorts;

Prediction 3: HLA class I loss diminishes antigen presentation, reducing T-cell recognition and effectiveness;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- HRR (hemoglobin, RDW): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HLA-A / HLA-B / HLA-C (class I antigen presentation): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.